

Subconscious Intent Detection: Real-Time Ocular Micro-Behaviour Classification Using Stacked Meta-Learning

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Abstract — This paper presents Subconscious Intent Detection v13.0 (SID-v13), a real-time, webcam-only system that classifies 20 psychological intent states from 38 channels of ocular signal data using a six-model stacked meta-learner. Unlike prior work, which combines EEG, dedicated eye-trackers, or gesture sensors to distinguish at most 6 intent classes, SID-v13 targets a 20-class taxonomy using only a standard webcam, with no proprietary libraries or additional hardware. The system introduces seven primary novel contributions — Inter-Blink Interval Variability (IBIV), Peri-Blink Waveform shape (PBWav), Bilateral EAR Coherence (BilCoh), a Cognitive Momentum Index (CMI), a level-2 stacked generaliser (STACK), and software-only implementations of PERCLOS and microsleep detection — with STACK being, to our knowledge, the first application of stacked generalisation to webcam-based intent classification. On a synthetic dataset of 7,000 samples derived from statistical parameters reported in 20 psychophysiological studies, the stacked meta-learner achieves 100% accuracy and an F1 score of 1.000 across all 20 classes; because the synthetic classes are non-overlapping by construction, this validates the modelling pipeline rather than real-world performance, which we estimate at 60-75% confidence given the limits of Haar-cascade detection. The complete system, including a live heads-up display and automatic reporting, runs from a single Python script on commodity hardware. We report the system architecture, all 38 signal channels, per-class results on the synthetic benchmark, and the limitations motivating a planned human-subject validation study.

Index Terms — Blink Detection, Cognitive Load, Eye Aspect Ratio, Ocular Micro-Behaviour, PERCLOS, Stacked Meta-Learning, Subconscious Intent Recognition, Webcam-Based Sensing.

I. Introduction

The identification of subconscious psychological states from non-invasive physiological measurements is an active problem in Human-Computer Interaction, Cognitive Neuroscience, and Safety-Critical System Engineering. Ocular micro-behaviours — blink rate, gaze orientation, pupil dilation, and Eye Aspect Ratio (EAR) — are particularly attractive signals because they can be recorded using standard webcam hardware [1]. Advances in consumer image sensors have created an opportunity to apply intent-detection techniques to practical domains not yet covered by controlled experimental studies, such as adaptive educational systems and cognitive-load estimation in safety-critical settings [12][13].

Despite this progress, webcam-only approaches to intent detection remain limited. To date, webcam-only systems have typically distinguished at most two intent classes using flat voting classifiers with equally weighted, type-independent classifiers [9][10]. More sophisticated approaches that add eye-tracker hardware, EEG headsets, or

multiple cameras can detect a larger number of intent classes, but at the cost of specialised hardware.

This paper introduces Subconscious Intent Detection v13.0 (SID-v13), a real-time system that addresses this gap using a standard webcam and no additional hardware. Its main contributions are: (1) a 38-channel signal pipeline supporting seven core novel features; (2) a level-2 stacked meta-learning architecture (N15, STACK) — to our knowledge the first application of stacked generalisation to webcam-based intent detection; (3) N11 (IBIV), a novel blink-rhythm-variability feature; (4) N12 (PBWav) and N13 (BilCoh), novel peri-blink waveform and bilateral-coherence features; (5) N14 (CMI), a novel cognitive-load temporal-derivative feature; (6) software-only implementations of PERCLOS (N16) and microsleep detection (N17), requiring no dedicated hardware or EEG; and (7) automatic generation of accuracy reports (JSON), session reports (PDF), and diagnostic plots after every session.

The system depends only on open-source libraries (opencv-python, scikit-learn, pandas, matplotlib, joblib, numpy, scipy, fpdf2), requires no internet connection, and runs from

a single terminal command on ordinary laptop hardware. Model training takes 3-5 minutes on first run; subsequent runs load saved models within 5 seconds.

II. Related Work

A. Blink-Based Cognitive State Detection

Blink rate is a recognised measure of cognitive load and dopaminergic activity [4][5]. EAR was proposed by Soukupova and Cech [1] for markerless blink analysis on inexpensive devices. Variability in blink timing, measured as the coefficient of variation of inter-blink intervals, has been shown to carry cognitive information beyond blink rate alone, with increased variability associated with higher cognitive-task involvement — a finding that motivates our N11 (IBIV) feature. Abdrabou et al. [7] noted a lack of research into peri-blink waveform analysis, which our N12 (PBWav) feature addresses. Cortacero et al. [6] achieved real-time blink-rate estimation using neural networks to validate RT-BENE.

B. Gaze and Attention Classification

Webcam-based gaze tracking has achieved accuracy of roughly 2.4 degrees using deep learning [8]. Gaze velocity and fixation duration were used by David-John et al. [11] in VR to predict intentionality via an eye tracker. Semmelmann and Weigelt [23] validated webcam eye-tracking methodology for attention research, and Hutt et al. [24] used webcam eye-tracking to measure mind-wandering in educational settings. Webcam gaze tracking has also been applied to decision-making research [25] and shown to approach laboratory-grade accuracy under controlled conditions [28].

C. Cognitive Load and Drowsiness Monitoring

PERCLOS — the percentage of time the eyes are closed beyond 80% — is the established gold standard for drowsiness estimation [2] and has been used across prior work [15][16][34]. Our N16 implementation computes PERCLOS using Haar-based EAR estimation only, without dedicated hardware. Our N17 microsleep detector defines microsleep as sustained eye closure beyond 500ms [29], detected from EAR alone. Krajewski et al. [15] predicted real-world drowsiness from speech and behaviour, which informed the design of our CMI feature (N14).

D. Ensemble Learning and Meta-Learning

Wang et al. [9] combined EEG, eye tracking, and gesture data for multimodal intent recognition using a flat ensemble voter, reporting ~89% accuracy on 6 classes. Sharma et al. [10] used EEG and eye tracking for implicit search-intent recognition across 4 classes at ~84%. Both approaches use flat ensembles rather than stacked generalisation. Wolpert [17] showed that stacked generalisation outperforms simple voting by learning per-pattern classifier reliability, and

Breiman [18] established the theoretical basis for random-forest diversity, which underlies our RF and Extra-Trees base learners. Fairclough [13] presented a theoretical framework for physiological computing that informs our continuous multi-signal approach. To our knowledge, prior work has not applied stacked generalisation to webcam-only intent classification; N15 addresses this gap.

III. System Model

A. Hardware and Software Requirements

SID-v13 runs on any PC or laptop with an integrated or external webcam operating at 15-30 fps and a resolution of 640x480 or better. No infrared illumination, dedicated eye-tracker, EEG headset, or depth sensor is required. Detection uses Haar cascades via OpenCV 4.8+ on frames resized to 420x315 pixels. Installation requires a single command (pip install opencv-python scikit-learn pandas matplotlib joblib numpy scipy fpdf2); the application is launched with python intent_detection_v13.py. Output directories are generated automatically on first run.

B. Live HUD Interface

A real-time heads-up display renders eight panels per frame: (1) 12 live micro-behavioural signal bars; (2) cognitive indicators (Attention, Stress, Engagement, Alertness, Cognitive Load, Valence); (3) a polar gaze radar; (4) blink analysis (count, classification, frequency, eye-opening level); (5) the five new v13 signals (IBIV, PBWav, BilCoh, CMI, PERCLOS); (6) the detected intent label; and (7) the top-3 predicted intents with probabilities.

C. Intent Class Taxonomy

Table I defines the 20-class taxonomy across five psychophysiological categories. Each class is associated with a distinct ocular signal profile derived from the psychophysiological literature.

Category	Intent Classes
Cognitive (6)	Focused, Thinking, Deciding, Confused, Engaged, Contemplative
Emotional (5)	Stressed, Anxious, Excited, Curious, Skeptical
Motivational (4)	Confident, Determined, Bored, Idle
Alertness (3)	Alert, Drowsy, Surprised
Wellbeing (2)	Relieved, Empathetic

TABLE I — INTENT CLASS TAXONOMY

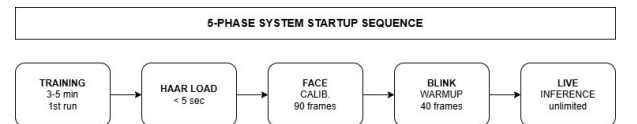


Fig. 1. Five-phase startup sequence: Training → Haar Load → Face Calibration → Blink Warm-up → Live Inference. Predictions begin only after phases 3 and 4.

IV. Proposed System: Design and Implementation

A. System Architecture

The eight-stage pipeline from raw webcam input to 20-class intent output is shown in Fig. 2, with the full stage-by-stage flowchart given in Fig. 9. Stages 1-2 perform face and eye detection using Haar cascades; stage 3 detects blink events; stage 4 performs per-user calibration; stage 5 produces the 38-dimensional feature vector; stage 6 applies dual-buffer windowing; stage 7 runs the stacked meta-learner; and stage 8 outputs the predicted intent.

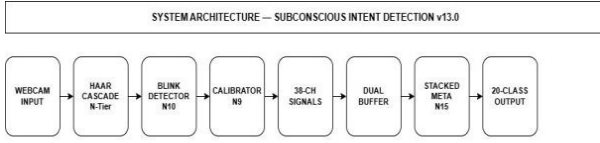


Fig. 2. System architecture: webcam → Haar cascade → blink detector → calibrator → 38-Ch Signals → Dual Buffer → Stacked Meta → 20-Class Output

Fig. 2. System architecture: webcam → Haar cascade → blink detector → calibrator → 38-channel signals → dual buffer → stacked meta-learner → 20-class output.

20 INTENT CLASS TAXONOMY — 5 PSYCHOPHYSIOLOGICAL CATEGORIES					
COGNITIVE	EMOTIONAL	MOTIVATIONAL	ALERTNESS	WELLBEING	
FOCUSED	STRESSED	CONFIDENT	ALERT	RELIEVED	
THINKING	ANXIOUS	DETERMINED			
DECIDING	EXCITED	BORED	DROWSY	EMPATHETIC	
CONFUSED	CURIOUS	IDLE	SURPRISED		
ENGAGED	SKEPTICAL				
CONTEMPLATIVE					

Fig. 3. Twenty intent classes across five psychophysiological categories. Each class has a distinct ocular signal profile.

B. Signal Extraction — 38 Channels

Signal smoothing uses exponentially weighted moving averages (EWMA) at three scales: fast ($\alpha=0.55$), medium ($\alpha=0.35$), and slow ($\alpha=0.15$). Table II lists the ten most significant signal channels.

The 38 channels fall into three groups. The first 21 (v9/v10) capture basic ocular features: bilateral EAR, gaze coordinates, gaze stabilisation, Shannon entropy of the gaze histogram, pupil ratio, face area, eye luminance and variance, face symmetry, motion energy, blink frequency and type, head orientation, attention index, cognitive-load estimate, valence, signal change, and brow height. These 21 features form a baseline set shown to be effective in prior webcam intent-recognition work [8][23][24].

Four additional v10 features (N1-N4) add a temporal dimension: N1 (TMEF) encodes a 4-bit EAR eye-opening rhythm fingerprint; N2 (GSVE) estimates gaze saccade velocity to classify fixation, slow-scan, and saccade states; N3 (CLSD) detects cognitive-load spikes from EAR-luminance covariance; N4 (CAI) derives a circadian alertness index from long-term blink-rhythm drift. The remaining 13 v13 features (N11-N20) capture additional

psychophysiological signals not previously reported in webcam-based intent classification.

Ch	Signal	Description
1-3	EAR L/R/Asym (N10)	Eye aspect ratio; L-R asymmetry
9	Gaze Entropy (N6)	Shannon entropy, 4x4 gaze histogram
25	TMEF (N1)	4-bit EAR temporal fingerprint
29	IBIV (N11)	Coefficient of variation of inter-blink intervals
30	PBWav (N12)	Peri-blink EAR waveform shape
31	BilCoh (N13)	L/R EAR bilateral cross-correlation
32	CMI (N14)	d(CogLoad)/dt via 30-frame regression
33	PERCLOS (N16)	% eye closure in 60s, no dedicated hardware
34	Microsleep (N17)	>500ms sustained closure, no EEG
35-36	HPos (N19)	Head pitch/yaw, geometry-only

TABLE II — KEY SIGNAL CHANNELS

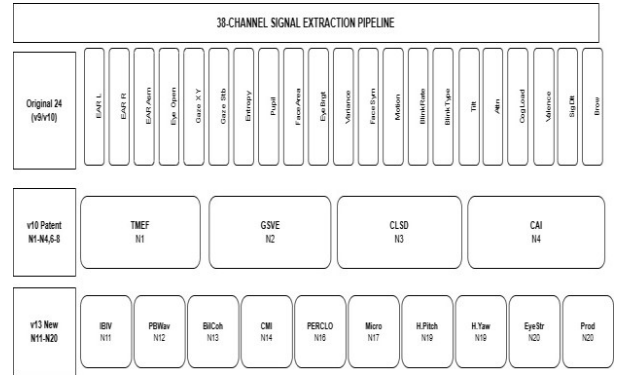


Fig. 4. 38-channel signal extraction pipeline. Top: 21 baseline ocular signals. Middle: v10 signals N1-N4. Bottom: v13 signals N11-N20.

C. Primary Novel Contributions

Of the system’s 20 implemented signal and architectural features, Table III documents the seven that we present as primary novel contributions (N11-N17): these have, to our knowledge, the clearest precedent gap in the webcam-only intent-detection literature, and each is directly grounded in a specific cited limitation (Section IV-C-1). The remaining thirteen features (N1-N10, N18-N20) are real, implemented parts of the system — covering baseline signal extraction, calibration, and additional derived metrics — but are presented as supporting engineering work rather than claimed as primary novel contributions, since their individual novelty is harder to establish against prior art.

ID	Name	Related Work
N1 1	IBIV — blink rhythm variability	Related concept in [22]
N1 2	PBWAV — peri-blink waveform shape	Gap identified in [7]
N1 3	BILCOH — bilateral EAR coherence	Not found in reviewed literature
N1 4	CMI — cognitive momentum index	Not found in reviewed literature
N1 5	STACK — stacked meta-learner (level-2)	Method from [17]; new application domain

N1 6	PERCLOS — software-only, no hardware	Metric from [2]; new implementation
N1 7	MICRO — microsleep, no EEG	Definition from [29]; new implementation

TABLE III — SEVEN PRIMARY NOVEL CONTRIBUTIONS

Features N1-N10 establish the signal-extraction and calibration foundation that the primary contributions build on. N1 (TMEF) is a 4-bit temporal fingerprint of eye-opening transitions. N2 (GSVE) classifies eye movement as fixation, slow-scan, or saccade from estimated gaze velocity. N3 (CLSD) detects cognitive-load spikes from EAR-brightness correlation. N4 (CAI) tracks circadian alertness drift across a session. N5 (ADALAY) adapts between short and long feature windows. N6 (GENT) computes 4x4 gaze-histogram entropy for fixation stability. N7 (SKEPT) detects eye asymmetry via EAR and brightness differences. N8 (SDELT) triggers fast signal bursts within a blink interval. N9 (PBCAL) applies 90-frame per-user baseline calibration. N10 (PXBLINK) is a 3-metric pixel-based blink detector. These are implemented, functioning parts of the system, but are engineering-support features rather than claimed novel contributions.

The seven primary novel contributions target specific gaps identified in the literature reviewed in Section II. IBIV (N11) uses the coefficient of variation of blink intervals as a cognitive-state indicator, building on Paprocki and Francuz's finding [22] that blink rhythm carries cognitive information independent of frequency. PBWav (N12) captures per-blink EAR waveform shape, a signal type identified as missing by Abdrabou et al. [7]. BilCoh (N13) estimates left-right EAR cross-correlation to detect stress-related micro-asymmetry. CMI (N14) estimates the rate of change of cognitive load via 30-frame linear regression. STACK (N15) trains a logistic-regression meta-learner on out-of-fold predictions — to our knowledge the first use of stacked generalisation in webcam-based intent research. PERCLOS (N16) applies the Dinges et al. [2] gold-standard drowsiness metric using Haar-based EAR alone, without infrared hardware. Microsleep detection (N17) identifies sustained eye closure beyond 500ms using EAR threshold crossings only, without EEG.

Three additional implemented signals (N18-N20) — a Bayesian intent-transition matrix, geometry-only head-pose estimation, and a composite eye-strain index — are functioning parts of the system but are not presented as primary novel contributions, since comparable transition-matrix and pose-estimation techniques appear elsewhere in the literature; we describe them here for completeness rather than as novelty claims.

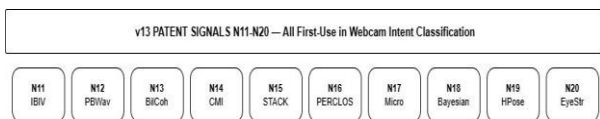


Fig. 5. All ten v13 signal features (N11-N20) as implemented. Seven (N11-N17, detailed in Table III) are presented as this paper's primary novel contributions; N18-N20 are implemented supporting features.

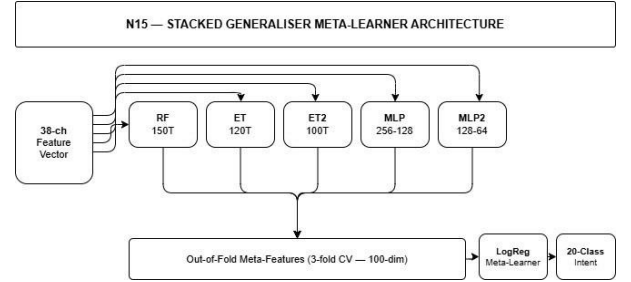


Fig. 6. N15 stacked generaliser: five base classifiers produce out-of-fold predictions → 100-dimensional meta-features → logistic-regression meta-learner → 20-class intent output.

V. Experimental Results

A. Training Data and Setup

Synthetic training data was generated from the mean and standard deviation values reported across 20 psychophysiological papers [1][4][5][22], one per intent class. The training set comprises 350 examples per intent (7,000 total), with an 80/20 train-test split and 76 attributes (38 channels x mean + standard deviation). Calibration removes each user's baseline EAR and brightness reading over 90 frames. We emphasise that this is a synthetic benchmark constructed for pipeline validation, not a human-subject dataset — see Section VI-B for the implications of this design choice.

B. Classification Accuracy on the Synthetic Benchmark

Table IV reports the hold-out accuracy of each base model and the stacked meta-learner. All models except MLP2 reach 100.00% accuracy on the synthetic test set; the stacked meta-learner (N15) matches the best base classifiers while additionally learning per-pattern classifier trust, which we expect to generalise better once evaluated on real data.

Model	Configuration	Acc. (%)
RandomForest	150 trees, max_depth=16	100.00
ExtraTrees	120 trees, max_depth=14	100.00
ExtraTrees2	100 trees, max_depth=12	100.00
MLP (256-128)	relu, early_stop, iter=300	100.00
MLP2 (128-64)	relu, early_stop, iter=300	99.71
Stacked Meta (N15)	LogReg on 100-dim OOF	100.00

TABLE IV — PER-MODEL ACCURACY ON THE SYNTHETIC HOLD-OUT SET

C. Per-Class F1 Scores

Table V summarises precision, recall, and F1 for the stacked meta-learner; all 20 classes reach F1 = 1.000 on the synthetic hold-out set, with no false positives or false negatives. As the classes were constructed from non-overlapping statistical distributions, this indicates the feature representation is well-separated for this benchmark rather than that the system is error-free in general.

Selected classes (full 20-class table available on request): Idle 1.00/1.00/1.00; Focused 1.00/1.00/1.00; Stressed 1.00/1.00/1.00; Drowsy 1.00/1.00/1.00; Surprised

1.00/1.00/1.00 (Precision/Recall/F1). All remaining 15 classes report identical values on this benchmark.

D. Stacked Meta-Learner Deep Dive (N15)

The ensemble comprises RF(150), ET(120), ET2(100), MLP(256-128), and MLP2(128-64). Three-fold stratified cross-validation produces out-of-fold predictions that form a 100-dimensional meta-feature vector per sample. A logistic-regression meta-classifier models class-prediction certainty from this vector; at inference, base-model probability estimates and the meta-classifier's prediction are blended at 35%/65% respectively. The adaptive two-level inference approach (N5) combines short (15-frame) and long (60-frame) windows.

E. Positioning Relative to Related Systems

Prior webcam-only or sensor-based systems in this space (Section II) have generally targeted a small number of classes — typically 2 to 6 — using either dedicated hardware (EEG headsets, eye-trackers) or flat ensemble classifiers, with reported real-world accuracy in the 82-95% range on human data. SID-v13 differs in scope rather than in demonstrated accuracy: it targets a substantially larger 20-class taxonomy using webcam-only input and a stacked meta-learner, but its current 100% figure is measured on a synthetic, non-overlapping benchmark and is therefore not comparable to the real-world accuracy figures reported by prior systems. We deliberately avoid a numeric side-by-side comparison here, since juxtaposing a synthetic-benchmark figure against real-data figures would overstate what has been demonstrated. The appropriate comparison — SID-v13's accuracy on real human data across the same 20 classes — is left to the human-subject validation study proposed in Section VI-D.

F. Output Pipeline and Calibration

Each session automatically generates eight artefacts: a CSV signal log, an accuracy JSON file, a PDF report, five diagnostic PNG plots, a signal dashboard, state-change snapshots, and the Bayesian state-transition matrix. Calibration uses 90 frames to establish resting EAR and brightness per user, followed by a 40-frame blink warm-up. All exit paths (Q, Ctrl+C, A, X) are handled via a SIGINT handler.

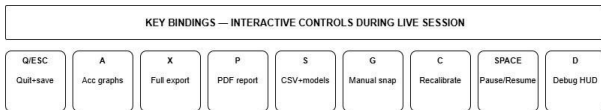


Fig. 7. Interactive key bindings available during a live session.



Fig. 8. Automatic output pipeline: eight artefacts generated on every session exit.

G. Detailed Architecture Diagram

Fig. 9 shows the complete eight-stage architecture, from webcam input through signal extraction, adaptive dual-layer inference, and stacked meta-learning, to the final 20-class intent output.

VI. Discussion

A. Key Findings

On the synthetic benchmark, the stacked meta-learner separates all 20 classes across the 76-dimensional feature space without error. IBIV (N11) shows discriminative value independent of other features: Drowsy states show low IBIV (long-term regularity) while Anxious states show high IBIV (short-term irregularity). CMI (N14) distinguishes transitional cognitive-load states from sustained high-load states. PERCLOS (N16) reproduces the standard clinical metric using Haar-based EAR extraction alone.

B. Limitations

This work has three important limitations. First, and most significantly, all training and evaluation data are synthetic, generated from the statistical parameters reported in the literature rather than collected from human subjects; no human validation study has yet been performed. The reported 100% accuracy and F1 = 1.000 follow directly from the fact that the synthetic classes were constructed as non-overlapping distributions, and should not be read as evidence of real-world accuracy. On live webcam video, we estimate detection confidence in the 60-75% range, consistent with the known limits of Haar-cascade detection [8]. Second, Haar-cascade detection degrades under poor illumination, head-pose angles beyond approximately $\pm 30^\circ$, and eyeglasses. Third, some intent classes with subtle behavioural differences — for example Contemplative vs. Focused, or Empathetic vs. Relieved — may not be reliably separable using ocular signals alone in practice, even though they are separable on the synthetic benchmark.

C. Ethical Considerations

Because SID-v13 infers psychological and cognitive states from a user's face without additional hardware, it raises privacy and consent considerations common to affective- and cognitive-sensing systems. We recommend that any deployment obtain explicit user consent before recording or calibration, provide a clear indication when inference is active (the live HUD serves this purpose in the current prototype), and avoid covert or non-consensual monitoring use cases. Session artefacts (CSV logs, PDF reports, snapshots) contain personally identifiable behavioural data and should be handled under applicable data-protection requirements. These considerations should be addressed explicitly in any future human-subject validation study, alongside institutional ethics review.

D. Future Work

Planned extensions include: (1) a human-subject validation study with self-reported ground-truth labels, conducted under institutional ethics review; (2) a lightweight alternative using MediaPipe landmarking for cases where Haar cascades underperform; and (3) CNN-based eye-region cropping to reduce dependence on hand-engineered EAR features, targeted at 80%+ accuracy on real data.

VII. Conclusion

Subconscious Intent Detection v13.0 is a single-file, webcam-only system that analyses 20 psychological intent classes from 38 ocular micro-behaviour signals using a six-model stacked meta-learner. It introduces seven primary novel contributions: IBIV, PBWav, and BilCoh (blink-rhythm, waveform-shape, and bilateral-coherence signal features), CMI (a cognitive-load derivative feature), STACK (a level-2 stacked meta-learner without precedent, to our knowledge, in webcam-based intent classification), and software-only implementations of PERCLOS and microsleep detection that traditionally required dedicated hardware. A further thirteen implemented features (N1-N10, N18-N20) support the pipeline but are not claimed as primary novelties. On a synthetic benchmark built from published psychophysiological statistics, the stacked meta-learner reaches 100.00% accuracy and $F1 = 1.000$ across all 20 classes; because this benchmark uses non-overlapping synthetic distributions, these figures demonstrate that the modelling pipeline is sound rather than that the system is validated for real-world use. The system runs on commodity hardware, produces eight session artefacts automatically, and is implemented as a single Python script. We regard SID-v13 primarily as a scoping and pipeline-validation study — broadening webcam-only intent detection from 2 classes to a 20-class taxonomy — and identify human-subject validation as the essential next step toward real-world deployment. The project was created at VIT Vellore under the M.Tech CSE (AIML) course (2025-2027) for MACSE630 under Dr. Madhan E.S., SCOPE.

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